

Nanostructure of the Deep Eutectic Solvents at Graphite Electrode Interface as a Function of Potential

Zhengfei Chen, Ben D McLean, Michael Ludwig, Ryan Stefanovic,
Gregory G. Warr, Grant Bruce Webber, Alister J. Page, and Rob Atkin

J. Phys. Chem. C, **Just Accepted Manuscript** • DOI: 10.1021/acs.jpcc.5b10624 • Publication Date (Web): 05 Jan 2016

Downloaded from <http://pubs.acs.org> on January 9, 2016

Just Accepted

"Just Accepted" manuscripts have been peer-reviewed and accepted for publication. They are posted online prior to technical editing, formatting for publication and author proofing. The American Chemical Society provides "Just Accepted" as a free service to the research community to expedite the dissemination of scientific material as soon as possible after acceptance. "Just Accepted" manuscripts appear in full in PDF format accompanied by an HTML abstract. "Just Accepted" manuscripts have been fully peer reviewed, but should not be considered the official version of record. They are accessible to all readers and citable by the Digital Object Identifier (DOI®). "Just Accepted" is an optional service offered to authors. Therefore, the "Just Accepted" Web site may not include all articles that will be published in the journal. After a manuscript is technically edited and formatted, it will be removed from the "Just Accepted" Web site and published as an ASAP article. Note that technical editing may introduce minor changes to the manuscript text and/or graphics which could affect content, and all legal disclaimers and ethical guidelines that apply to the journal pertain. ACS cannot be held responsible for errors or consequences arising from the use of information contained in these "Just Accepted" manuscripts.



**Nanostructure of Deep Eutectic Solvents at Graphite Electrode
Interfaces as a Function of Potential**

Zhengfei Chen,¹ Ben D. McLean,¹ Michael Ludwig,¹ Ryan Stefanovic,¹ Greg G. Warr,²
Grant B. Webber,¹ Alister J. Page,¹ Rob Atkin*,¹

¹Newcastle Institute for Energy and Resources, The University of Newcastle, Australia

²School of Chemistry, The University of Sydney, Australia

*rob.atkin@newcastle.edu.au

Abstract

Atomic force microscopy (AFM), density functional theory (DFT) calculations and contact angle measurements have been used to investigate the liquid – highly ordered pyrolytic graphite (HOPG) electrode interface for three deep eutectic solvent (DES) as function of applied potential. The DESs examined are 1:2 mixtures of choline chloride and urea (ChCl:urea), choline chloride and ethylene glycol (ChCl:ethylene glycol) and choline chloride and glycerol (ChCl:glycerol). DFT calculations reveal that in all cases the molecular component is excluded from the graphite interface at all potentials, while chloride and choline are attracted into the Stern layer at positive and negative potentials respectively. AFM force curves confirm these trends, and also show that the first near surface liquid layer in contact with the Stern layer is rich in the molecular component. The extent of near surface layering increases with potential and the hydrogen bonding capacity of the molecular component. The variation in the macroscopic contact angle with potential is consistent with changes in the Stern layer composition.

Introduction

Deep eutectic solvents (DESs) are formed by mixing an inorganic salt with a molecular hydrogen bond donor. The most commonly reported DESs to date consist of quaternary ammonium halide salts mixed with amide, carboxylic acid or hydroxyl hydrogen bond donors.¹⁻² The archetypal example in this respect is the choline chloride (ChCl):urea DES. ChCl rapidly complexes urea under mild conditions, and the eutectic point of the ChCl:urea mixture is at a 1:2 molar ratio for which the commonly reported melting point is 12 °C.¹

Abbott and co-workers originally proposed that freezing point depression in DESs is a consequence of the lattice energies of the inorganic salt and hydrogen bond donor, and the strength of the anion – hydrogen bond donor interaction.² This hypothesis has been confirmed for eutectic mixtures of choline chloride (ChCl) and urea.³ For this DES, (1) Cl⁻ – urea interactions overcome Ch⁺ – Cl⁻ lattice interactions for urea concentrations above 67 mol%, and (2) Cl⁻ – urea interactions are stronger than Ch⁺ – urea interactions. These two factors make the intercalation of urea into the solid ChCl lattice structure thermodynamically, and hence entropically, favoured. Consequently, Ch⁺ – Cl⁻ attractions are weakened in the mixture, lowering the lattice energy and the freezing point.

DESs share some physicochemical properties with conventional ionic liquids (ILs), such as low volatility, high thermal stability, high ionic conductivity, and non-flammability. In addition, DESs are less water sensitive than conventional ILs and can be produced by simple mixing of readily available reagents of low environmental impact.⁴⁻⁶ For these reasons, DESs have attracted increased attention¹ in fields as diverse as nanomaterial synthesis,⁷⁻⁸ catalysis,⁹ and surfactant self-assembly¹⁰ (see also several recent reviews^{5, 11-12}). DES are attracting most research interest for use as electrochemical solvents.^{4, 13-18} In this respect, the structure and composition of the DES at an electrified (charged) electrode surface is critical to its utility, as

per ILs where the interfacial nanostructure has been shown to be a key determinant of physical and electrochemical processes.¹⁹⁻²¹

In ILs, the structure of the liquid interface for graphite, silica, mica and gold electrode surfaces has been characterized previously.²¹⁻²⁸ Three regions have been identified at IL-solid interfaces: (1) the interfacial or Stern layer, (2) the transition or near-surface zone (the region over which the pronounced interfacial layer structure decays into the bulk) and (3) the bulk liquid.²⁴ As the electrode potential is varied, the ratio of cations to anions in the IL Stern layer changes in response, with cations preferentially adsorbed at negative potentials and vice versa.^{19, 21, 29} As the Stern layer templates structure in the near surface region, changes in potential also affect the near surface structure.^{24-25, 27} Thus, the potential dependent interfacial behaviour of ILs is a critical design parameter for their use as electrochemical solvents.

DES interfaces are inherently more complicated than IL interfaces. Whereas ILs consist of cations and anions only, DESs comprise cations, anions and a molecular component which, for the systems studied here, is present at twice the concentration of the parent salt. In some respects, DES can be conceptualized as concentrated solutions of salt in the molecular component, although speciation will play a key role; we will address the effect of speciation on bulk properties in a forthcoming manuscript. While electrochemical measurements have been used to draw inferences about the structure of DES double layers at platinum, gold, glassy carbon and mercury surfaces,^{14, 30} the interfacial nanostructure of DESs at an electrode surface has not previously been directly probed. Here we investigate the nanostructure of three common DESs – ChCl:urea, ChCl:ethylene glycol and ChCl:glycerol – on highly oriented pyrolytic graphitic (HOPG) electrodes as a function of potential using nanoscale atomic force microscope (AFM) force curves and macroscale electrowetting experiments, combined with quantum chemical simulations. The primary questions we address in this manuscript are:

1. Do the ChCl:urea, ChCl:ethylene glycol and ChCl:glycerol graphite interfaces have multilayer nanostructure like ILs?
2. How does the interfacial nanostructure differ as the molecular component is changed?
3. Does the choline, chloride or molecular component preferentially adsorb to the graphite Stern layer and, if so, how does this vary with potential?
4. Can changes in Stern layer composition be correlated with macroscopic contact angle measurements?

Experimental and Computational Methods

Deep Eutectic Solvent Preparation. ChCl (98 %, Sigma Aldrich), urea (99.5 %, Sigma Aldrich), ethylene glycol (99.5 %, BDH Chemicals) and glycerol (99.5 %, Thermal Fisher) were used without further purification. Each DES was prepared in the molar ratio of 1:2 ChCl:X (X = urea, ethylene glycol, glycerol) by stirring the mixture in a seal glass vial at 60 °C until a homogenous and clear solution was formed. The water contents for the ChCl:urea, ChCl:ethylene glycol and ChCl:glycerol DESs determined by Karl Fisher titration were 0.08 wt%, 0.04 wt% and 0.07 wt%, respectively.

Atomic Force Microscopy. Normal force measurements were performed on a Nanoscope 5 Multimode 8 atomic force microscope (AFM, Bruker Instruments) in contact mode. Standard silicon tips with a spring constant approximately 1 N/m (SPM Probe, MikroMasch, USA) were used throughout the experiments and they were cleaned with Milli-Q water and ethanol and then irradiated with ultraviolet light for 10 min prior to use.

The structure normal to the DES – HOPG interface was measured between ± 1 V using an AFM electrochemical cell described previously.²⁴ All AFM measurements were therefore made inside the electrochemical window of these DESs.¹⁷ The AFM cell uses a Cu counter electrode and Pt pseudo-reference electrode, with the HOPG substrate acting as the working

electrode. HOPG surfaces were freshly cleaved before each measurement. Potential was applied using an Autolab potentiostat, and the system was allowed to reach equilibrium at each potential for 10 minutes. AFM force curves were collected at a ramp size of 20 nm and a ramp rate of 0.5 Hz at room temperature ($\sim 22 \pm 1$ °C). More than 20 force curves were collected for each DES system at applied potentials of -1 V, -0.5 V, 0 V, +0.5, +1 V and open circuit potential (OCP), to ensure reproducibility. The most representative force curve for each system is discussed below. The OCP (vs Pt) for the ChCl:urea, ChCl:ethylene glycol and ChCl:glycerol DESs are +0.46 V, +0.35 V and +0.3 V, respectively. The positive potential for these ChCl based DESs at HOPG electrode is consistent with the recent report of ChCl:ethylene glycol on a glassy carbon electrode (+0.25 V vs Zn).³¹ The positive OCP means that the innermost layer must be anion rich, due to electrostatic interactions. IR adsorption studies at the ChCl:ethylene glycol – carbon interface at OCP reveal that the innermost layer also contains both Ch^+ cations and ethylene glycol molecules.³¹ Since both ChCl:urea and ChCl:glycerol DESs have positive OCP it is expected that the innermost layer for these DESs will also be enriched in Cl^- anions.

Electrowetting Experiments. Electrowetting experiments on HOPG were conducted on an OCA-20 pendent drop tensiometer (DataPhysics Instruments, Filderstadt, Germany) at room temperature (~ 22 °C). The potential was controlled using an Autolab potentiostat and the contact angle was calculated with a circular fitting method using the OCA software.

Quantum Chemical Calculations. The interactions between HOPG and chloride, choline, urea, ethylene glycol and glycerol were analyzed using density functional theory (DFT). Truhlar's M06-2X functional³² was employed in conjunction with a 6-31G(d) basis set. The M06-2X functional has been utilized previously to accurately describe ILs both in the bulk and adsorbed on HOPG.^{27, 33-34} Adsorption free energies of component moieties on a

model HOPG surface have been calculated at 298 K using harmonic vibrational frequencies.

Adsorption free energy ΔG_{ads} of a species X is defined here as:

$$\Delta G_{\text{ads}} = \Delta G(X + \text{HOPG}) - [\Delta G(X) + \Delta G(\text{HOPG})] \quad (1)$$

where G denotes the free energy of each individual species. Thus, negative ΔG_{ads} correspond to exoergic processes, while positive ΔG_{ads} correspond to endoergic processes. It should be noted that by this definition the adsorption free energy of species X is not influenced by the presence, or otherwise, of other species. In this way ΔG_{ads} for Ch^+ and Cl^- are the same in each DES. A hexagonal $\text{C}_{54}\text{H}_{18}$ graphene flake, *ca.* 1.5 nm in diameter, was used to approximate the HOPG surface. Applied potentials were approximated *via* positive and negative static dipole fields applied normal to the model HOPG surface. All calculations were performed using the Gaussian09 software package.²⁷

Results

Key physicochemical properties and structures for the $\text{ChCl}:\text{urea}$, $\text{ChCl}:\text{ethylene glycol}$ and $\text{ChCl}:\text{glycerol}$ DESs are given in Table 1. The freezing point of the DESs is strongly affected by the hydrogen bond donor type. The freezing point for the $\text{ChCl}:\text{urea}$, $\text{ChCl}:\text{ethylene glycol}$ and $\text{ChCl}:\text{glycerol}$ DESs are depressed by 121, 53 and 58 °C, respectively, relative to the freezing (melting point) of the corresponding hydrogen bond donor. The electrochemical windows for these DES are approximately 2V, as shown in Table 1, and the approximate size and structure of each DES component are shown in Figure 1. Dimensions were obtained by measuring distances between appropriate atoms of structures determined using Gaussian.²⁷

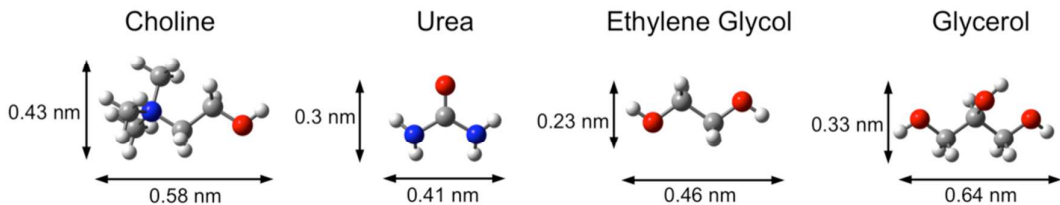


Figure 1. Approximate dimensions of DES component moieties. Atoms are: carbon (grey), nitrogen (blue), oxygen (red) and hydrogen (white). The Cl⁻ ion diameter is *ca.* 0.36 nm.³⁵

Table 1. Density, melting point and surface tension for the ChCl:urea, ChCl:ethylene glycol and ChCl:glycerol DESs. Unless otherwise stated all values were measured at 20 °C.

DES	Water Content (Wt%)	Density (g/cm ³)	Melting Point (°C)	Electrochemical window (V)	DES/air Surface Tension, γ (mN/m)
ChCl:urea	0.08	1.25 at 25 °C ³⁶	12 ¹	-1.2 to +1.1 ¹⁷	52 at 25 °C ³⁶
ChCl:ethylene glycol	0.04	1.12 ³⁷	-66 ³⁸	-1.2 to +1.2 ⁴	49 ³⁷
ChCl:glycerol	0.07	1.18 ³⁷	-36 ³⁸	-1.3 to +1.2 ⁴	56 ³⁷

In our previous AFM force curve studies of ILs on electrode surfaces differences in the width of the step closest to the surface enabled the Stern layer composition to be commented upon.^{19, 24-25} For DESs, the presence of the molecular component complicates analysis, as variation in step size with potential could be due to the populations of cations and anions or the molecular component changing. To enable interpretation, adsorption free energies (ΔG_{ads}) for the cation, anion and each molecular component on HOPG have been calculated as a function of potential using M06-2X/6-31G(d).

In the absence of applied potential, ΔG_{ads} is positive for all the molecular components (+13.6 (urea), +8.0 (ethylene glycol) and +13.7 (glycerol) kJ/mol) meaning the molecular

components are weakly repelled from the HOPG surface. This is a consequence of the mismatch between hydrophobicity of the graphite surface and the polarity of the molecular component. These ΔG_{ads} values varied only slightly (< 8 kJ/mol) as the potential was changed, consistent with the molecular hydrogen bond donors being electrically neutral. Thus, at all potentials between +1V and -1V the molecular component is repelled from HOPG.

Figure 2 shows ΔG_{ads} for Ch^+ and Cl^- ions on a HOPG surface as a function of applied potential. As expected, at negative potentials Ch^+ ΔG_{ads} values are large and negative (ΔG_{ads} up to -91 kJ/mol at -1 V), indicating strong adsorption, while the chloride ions (ΔG_{ads} up to +106 kJ/mol at -1 V) are repelled from the HOPG. This means the Stern layer at -1 V will be highly enriched in Ch^+ . At -1 V, Ch^+ is more strongly attracted to the HOPG surface than chloride is repelled. This is a consequence of dispersive interactions between Ch^+ and HOPG being stronger than between chloride and HOPG, and also the presence of cation – π interactions between the Ch^+ and the HOPG substrate. Cation – π interactions with aromatic solutes/solid substrates also play a key structural role in conventional ILs.³⁹⁻⁴² This difference persists at all potentials because the electrostatic attraction or repulsion adds to this dispersive component.

As the potential is made more positive, the attractive ΔG_{ads} for Ch^+ and repulsive ΔG_{ads} for chloride decrease, meaning that the degree of cation enrichment will decrease. For both ions, the change in ΔG_{ads} with potential is close to linear and the absolute value of the gradient is approximately equal to Faraday's constant consistent with the Nernst equation. Deviations from linearity are strongest at 0 V as in the absence of potential dispersion rather than electrostatic interactions dominate the interactions.

Between 0 V and $\sim +0.5$ V ΔG_{ads} for both ions is negative and similar, so the Stern layer concentration of cations and anions will be more closely matched. The lines joining the ΔG_{ads} values for the cation and anion in Figure 2 intersect at +0.35 V. Equal adsorption energies mean that the populations of cations and anions in the Stern layer will be equal, and this potential therefore predicts the point of zero charge (PZC) to a first approximation. Mapping the position of the PZC onto the experimental data is non-trivial as 0 V is not known in absolute terms when a quasi-reference electrode is used (as is required for the small volume (~ 0.3 mL) AFM cell). This will be discussed below. When the potential is increased above +0.35 V, the Stern layer begins to become increasingly enriched in chloride.

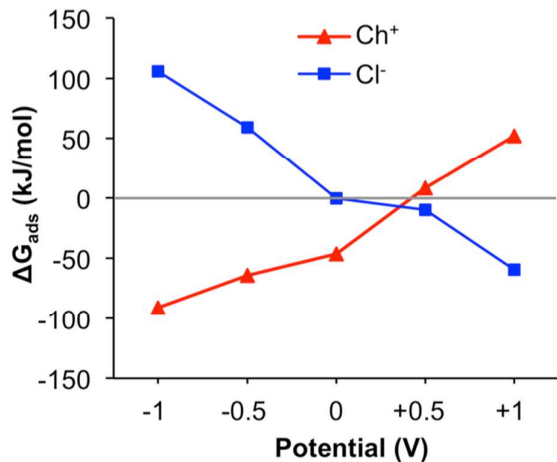


Figure 2. M06-2X/6-31G(d) ΔG_{ads} values (kJ/mol) for choline and chloride.

AFM force curves for pure glycerol and ethylene glycol are presented in Figure 3. Corresponding data for pure urea was not obtained due to its high melting point. Force curves as a function of potential for the ChCl:urea, ChCl:ethylene glycol and ChCl:glycerol DESs are presented in Figures 4, 5 and 6 respectively. For each system and potential the force curves shown were chosen by ascertaining the mean step number and push-through forces for

more than 60 force curves collected over different days. This enabled a representative force curve to be selected that most closely matched the average values. Presentation of a single force curve is favoured over an average⁴³ because averages obscure features that, although consistent, occur at small normal forces or slightly different separations, such as the small steps at wide separations.

In an AFM experiment the absolute tip – sample separation distance is not known. The zero position in force curves is determined from the region in the raw deflection data when the tip and the surface are in constant compliance. Therefore, a separation of zero does not mean that the tip is *necessarily* in contact with the surface, as constant compliance also occurs when the tip is presses against matter that it cannot displace.

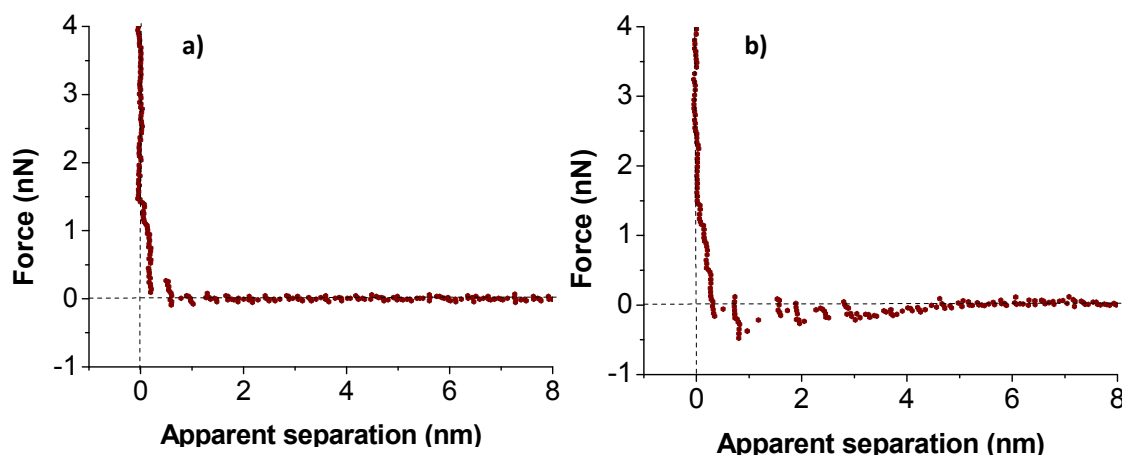


Figure 3. Normal force distance profiles for pure (a) ethylene glycol and (b) glycerol without applied potential on graphite.

For pure glycerol, ethylene glycol, and the DESs at all potentials, at least one step is noted in the force data. Steps result from interfacial liquid layers being expelled from the space between the substrate and the AFM tip. The “push-through” force required to rupture a layer is a function of the strength of cohesive interactions between layers, or of the layer and the

surface if the tip pushes through to the final layer. Similar steps are noted in ILs,²⁴⁻²⁵ and for molecular liquids due to solvation layers.⁴⁴ The difference between IL interfacial structure and molecular liquid solvation layers is that IL ions have preferred orientations or structures (i.e. are more precisely arranged) than the molecules in molecular liquid solvation layers. As a result, more and better formed layers are detected for ILs than molecular liquids.⁴⁵ Additionally, interfacial layering becomes significantly more pronounced for ILs as the potential is increased or decreased,²⁴ whereas no change occurs for non-conducting molecular liquids. (In fact, for molecular liquids in the absence of added electrolyte a surface potential cannot be measured).

The force curves for pure glycerol and pure ethylene glycol reveal a few clear near surface oscillations followed by a final, narrow and steep compression before constant compliance is achieved at zero apparent separation. The oscillations are superimposed on a weak van der Waals attraction in pure glycerol. Given the positive ΔG_{ads} for both glycerol and ethylene glycol on HOPG, it is probable that for these liquids the tip is in contact with the surface at zero separation. This is also suggested by the shape of the final repulsive part of the force curve, and specifically by its compressibility and absence of a distinct push-through threshold for the AFM tip to jump into contact with the surface, which would be expected if the intervening molecular layer adhered to the substrate.

The average packing dimension⁴⁶ determined from the liquid density has proven to be a remarkably accurate predictor of step (oscillation) period for both molecular liquids and ILs. For glycerol and ethylene glycol, this is calculated to be 0.50 nm and 0.45 nm, respectively, which are measurably larger than the final layer thicknesses of 0.3 nm for glycerol and 0.2 nm for ethylene glycol. However, these small step sizes are consistent with the expected thickness of a layer of glycerol or ethylene glycol molecules lying along the surface in a flat orientation (see Figure 1). Similar results have been obtained previously for linear alkanes

1
2
3 and n-dodecanol,^{44, 47} although AFM alone cannot distinguish between surface and
4
5 confinement effects in this case. IR adsorption measurements have previously shown the
6
7 ethylene glycol to adsorb to glassy carbon in a mixed morphology.³¹ The periods of
8
9 subsequent, near surface oscillations are larger and consistent with expectations from the
10
11 average packing dimension, meaning that molecules in these layers do not have preferred
12
13 orientations.
14

15
16
17 The force curves for the three DESs (Figures 4-6) are qualitatively different from those of
18
19 the pure molecular components. All exhibit step-like oscillations with negligible underlying
20
21 van der Waals attractions, and a near-vertical (nearly incompressible) repulsive wall
22
23 immediately before the jump to zero apparent separation. At each potential the number of
24
25 oscillatory steps increases upon changing molecular component from urea to ethylene glycol
26
27 to glycerol, and for each DES increasing the magnitude of the potential also increases the
28
29 number of steps (most obviously for ChCl:glycerol).²⁴⁻²⁵ The first of these is a consequence
30
31 of a range of factors including packing considerations, the liquid dielectric constant and
32
33 cohesion between layers, amongst others.²⁴ The effect of potential is similar to that seen on
34
35 near surface layering ILs, and occurs because as the Stern layer becomes more enriched in
36
37 counterions this templates better defined structure in near surface layers.²⁴⁻²⁵ However, in
38
39 DESs the situation is complicated by the presence of the molecular component.
40
41
42
43

44 The three component complexity of DESs means that step widths cannot be used to
45
46 unambiguously ascertain the composition of layers as is done in ILs, with the exception of the
47
48 step nearest the surface.
49
50
51
52
53
54
55
56
57
58
59
60

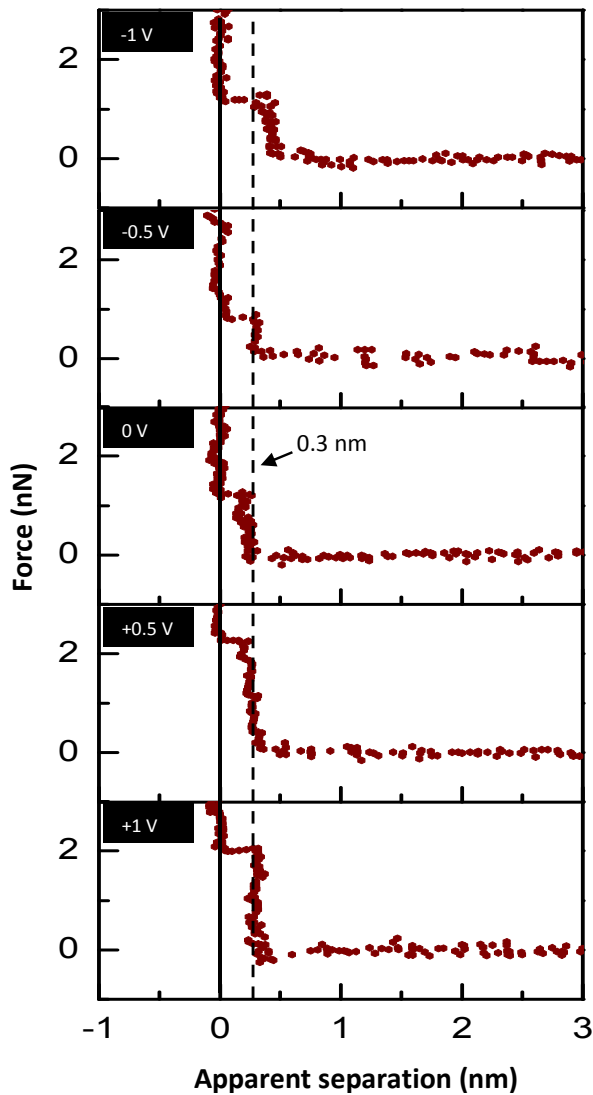


Figure 4. Normal force distance profiles for ChCl:urea as a function of potential.

The thickness of the final jump in the force curve for both ChCl:urea and ChCl:ethylene glycol is consistently 0.3 nm at all potentials between +1 V and -0.5 V. This is too small to ascribe to displacement of either Ch^+ (Figure 1) or Cl^- , which has a diameter of 0.35 nm, but is consistent with the width of an aligned layer of the molecular components, urea (0.3 nm) and ethylene glycol (0.23 nm), and also with the final step width of the pure ethylene glycol force curve. For ChCl:glycerol the corresponding innermost step thickness is 0.4 nm at all

potentials. This larger step size is consistent with the greater width of the aligned glycerol. It is thus concluded that the final step in the DES force curves is due to displacement of the molecular component-rich, penultimate near surface layer, which is in contact with an underlying Stern layer that it cannot displace. The step widths suggest that the molecular components are preferentially oriented parallel to the substrate. Zero apparent separation thus corresponds to the AFM tip pushing against an ion rich Stern layer that it cannot displace. This interpretation is consistent with the calculated ΔG_{ads} for Ch^+ and Cl^- that are similar in magnitude to structurally similar IL ions, and which AFM imaging and nanotribology measurements have definitively showed could not be displaced from graphite at comparably high forces.⁴⁸⁻⁵¹ All of the DESs contain trace amounts of water. Water is strongly repelled by the graphite interface which results in a contact angle of around 90° , and calculations have confirmed positive adsorption energies at close separations.⁵²⁻⁵³ Given the affinity of the DES cations and anions for the graphite surface shown in Figure 1, it is thus highly unlikely any trace water is concentrated in the Stern layer, and there is similarly no reason why water would be concentrated in the near surface layer. As the affinity of cations and anions for the graphite increases as the potential is decreased and increased respectively, the possibility of water being incorporated into the Stern layer at applied potentials is even lower. It is thus concluded that trace water is essentially absent from the Stern layer, and likely has approximately the same concentration in near surface layers as in the bulk.

When the potential is decreased to -1 V, the final step width increases to 0.45 nm for all three systems. At such negative potentials, the Stern layer is highly enriched in Ch^+ . Compared to lower potentials, here the Ch^+ ions will prefer an upright conformation to facilitate dense packing of quaternary ammonium groups close to the electrode surface. The increase in near surface layer step width could be a consequence of this Ch^+ conformation change altering packing of the molecular component in the penultimate surface layer, or that

a second, cation-rich layer is required to satiate the surface potential, similar to the crowding effects seen in many IL systems.⁵⁴ Corresponding effects are not observed at positive potentials in chloride-rich Stern layers because the small, symmetrical chloride ion can pack tightly onto the surface without any such conformational change. At higher potentials, chloride crowding at the surface may occur, but out of the accessible range for these DESs, as +1 V is near the positive potential limit.

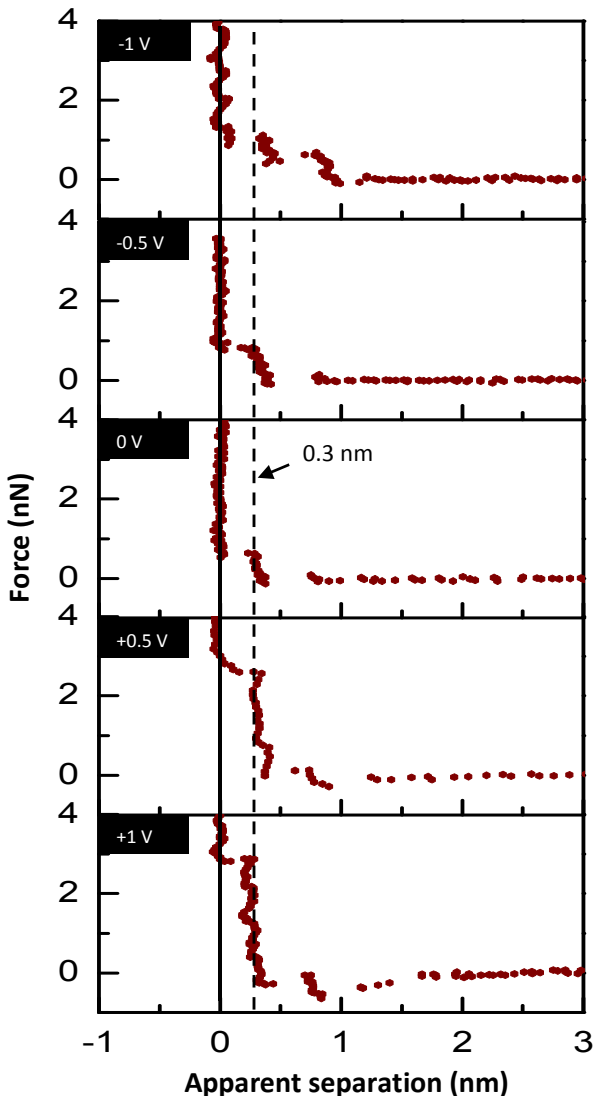


Figure 5. Normal force distance profiles for ChCl:ethylene glycol as a function of potential.

The magnitude of the final layer push through force is primarily a consequence of cohesive interactions between the Stern layer and near surface layer.⁵⁵ Figure 7 presents the final layer push through force at each potential for each system. Although different cantilevers and tips were used for different systems, the spring constants and tip shapes were very similar so the forces should be directly comparable. The force curves for the data points between 0 and 0.5 V are shown in the Supporting Information.

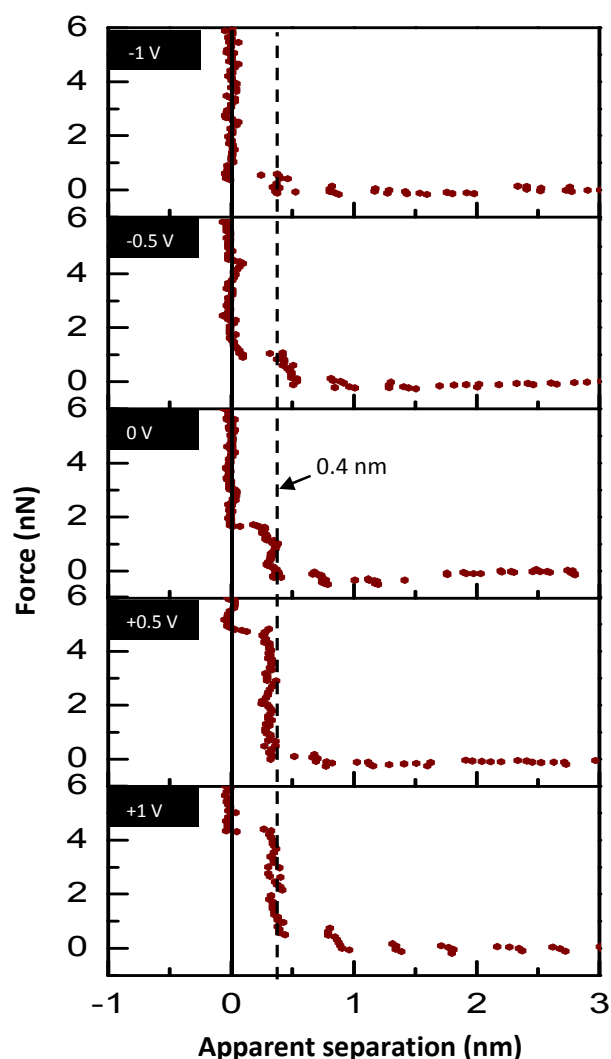


Figure 6. Normal force distance profiles for ChCl:glycerol as a function of potential.

The rupture force is small and constant at potentials below about +0.2V, and identical (within experimental error) for all three systems. There is then a sharp increase to a new plateau force at more positive potentials, which varies between systems, with the limiting rupture force falling in the order glycerol > ethylene glycol > urea. The transition point also differs slightly between systems, lying between 0.3 V and 0.5 V for ChCl:urea, 0.25 V and 0.5 V for ChCl:ethylene glycol, and 0 V and 0.3 V for ChCl:glycerol.

This behaviour reflects the changing composition of the underlying and strongly bound ionic Stern layer, and the transition potential is consistent with expectations based on calculated adsorption free energies of Ch^+ and Cl^- (c.f. Figure 2). At positive potentials the molecular component rich penultimate surface layer is interacting with a well ordered, chloride rich Stern layer, which packs neatly onto the atomically smooth HOPG surface. This facilitates hydrogen bonding between chloride and the molecular component, which leads to high push through forces that are sensitive to the structure and H-bond capacity of the molecular component. At negative potentials however, this molecular layer interacts with a Ch^+ rich Stern layer. Here H-bonding is expected to be less efficient due to the reduced density, order, and specific orientation of the Ch^+ ion layer, including transverse Ch^+ rotations. This means that cohesive interactions between the near surface layer are weaker with Ch^+ than chloride. The rupture force thus decreases and is less sensitive to the nature of the molecular component. Additionally, hydrogen bonds between chloride and a donor are ~ 3 times stronger than between hydroxyl and hydrogen bond donors,⁵⁶ which will also lead to stronger cohesive interactions between the near surface layer and chloride than choline. It is not currently possible to resolve which of these effects is of greater consequence for the reduction in rupture force.

As the Stern layer changes composition from cation-rich to anion-rich in the potential region where the rupture force changes sharply, the PZC for these systems must lie between

0.3 V and 0.5 V for ChCl:urea, 0.25 V and 0.5 V for ChCl:ethylene glycol, and 0 V and 0.3 V for ChCl:glycerol. This is consistent with the PZC expected from calculated ΔG_{ads} values, which are equal for Ch^+ and Cl^- at +0.35 V. Measured differences between DESs likely reflect the interactions between ionic and molecular components not accounted for in the calculations.

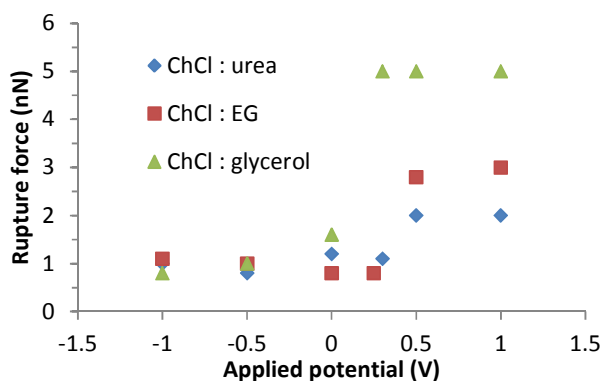


Figure 7. Inner layer rupture force for various DES on HOPG as a function of applied potential.

The contact angles ChCl:urea, ChCl:ethylene glycol and ChCl:glycerol DESs on graphite as function of potential are shown in Figure 8. The contact angle of a sessile ionic liquid droplet on a solid electrode typically has a maximum at the PZC, decreases gently as the potential is made more positive, and decreases rapidly as the potential is made more negative, a phenomenon referred to as electrowetting.⁵⁷ The electrowetting behaviour of these DESs is consistent these expectations, and with PZC's determined from AFM data, although the contact angle maximum is extremely broad: contact angles measured between 0 V and 0.5 V (not shown) all lay within 1° , so could not been used to independently identify the PZC.

The equilibrium contact angle (θ_0) is a function of the various interfacial tensions:

$$\cos \theta_0 = (\gamma_{SV} - \gamma_{SL})/\gamma \tag{1}$$

where θ_0 is angle, γ_{SV} is the solid/vapor interfacial tension, γ_{SL} is the solid/liquid interfacial tension and γ is the liquid/vapor interfacial tension.⁵⁷ As γ_{SV} and γ remain constant during electrowetting any change in θ_0 must be due to γ_{SL} changing.⁵⁷ The saturated contact angle of a sessile liquid drop on a solid electrode surface can be estimated as⁵⁷⁻⁵⁸ $\cos \theta_S = \gamma_{SV}/\gamma$, since at this limit $\gamma_{SL} = 0$. The surface tension of a graphene sheet is 46.7 mN/m,⁵⁹ therefore using the values in Table 1 we obtain $\theta_S = 26^\circ$, 13° and 34° for ChCl:urea, ChCl:ethylene glycol and ChCl:glycerol, respectively. Thus the contact angle at -1 V (the edge of the electrochemical window) still has not reached its saturation limit. Nonetheless, within the electrochemical window, the trends in electrowetting are in keeping with changes in Stern layer composition determined from the AFM force curves. The contact angle decreases markedly as the Stern layer becomes enriched in Ch^+ which decreases cohesive attractions between the Stern layer and near surface layer.

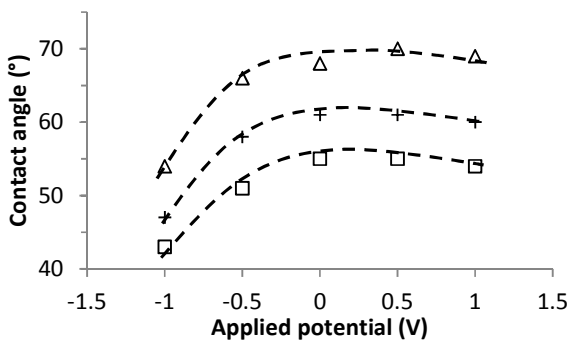


Figure 8. Contact angles of choline based DESs on graphite as function of potential; ChCl:urea (+); ChCl:ethylene glycol (□); and ChCl:glycerol (△).

Conclusions

The nanostructure of the interfaces of HOPG with ChCl:urea, ChCl:ethylene glycol and ChCl:glycerol have been investigated using DFT calculations, AFM force curves, and contact angle measurements. DFT calculations show that the molecular component is repelled from the surface at all potentials and, as expected, Ch^+ is attracted to the surface at negative potentials and chloride at positive potentials. AFM force curves have a step-like appearance due to the tip rupturing near surface liquid layers. Strong interactions between the ions and HOPG mean that the AFM tip is unable to rupture the Stern layer to move in to contact with the substrate, and the width of the final measured step reveals the near surface layer is rich in the molecular component. The force required to rupture this final layer is determined by the strength of interactions between molecular components in the near surface layer and Stern layer ions. As such, variation in the magnitude of the rupture force can be used to make inferences about the Stern layer composition and the PZC.

The force curves show that the three DESs have multilayer nanostructure, but generally the number of layers and push through forces are less than what is seen for ILs. This could be due to a variety of factors, but perhaps most importantly the presence of the molecular component in addition to cations and anions allows efficient packing without layering. The extent of layering increases as the molecular component hydrogen bonding capacity is increased due to stronger cohesive interactions between adjacent layers. The extent of near surface layering increased with the magnitude of the potential due more precise ion packing in the Stern layer leading to better defined packing of the molecular component in the near surface layer. Broadly speaking, changes in the Stern layer composition can be correlated with the variation in the macroscopic contact angle with potential.

To our knowledge, this is the first time the interfacial nanostructure of DESs has been probed at an electrode surface as a function of potential. Thus, these results are an important

step towards developing a detailed understanding of the DES electrical double layer, which is of key importance for the use of these solvents in electrochemical systems.

Supporting Information

Additional normal force curves for the DES's at applied voltages between 0 and +0.5 V are available electronically from <http://pubs.acs.org/journal/jpcceck>.

Corresponding Author

*E-mail: rob.atkin@newcastle.edu.au.

Acknowledgements

The authors acknowledge helpful conversations with Dr Hua Li (University of Newcastle) concerning electrowetting. AJP, GGW and RA acknowledge support from Australian Research Council Discovery Projects, and RA acknowledges ARC for a Future Fellowship. This research was undertaken with the assistance of resources provided at the NCI National Facility systems at the Australian National University and INTERSECT systems, through the National Computational Merit Allocation Scheme supported by the Australian Government.

References

- (1) Abbott, A. P.; Capper, G.; Davies, D. L.; Rasheed, R. K.; Tambyrajah, V. Novel Solvent Properties of Choline Chloride/urea Mixtures. *Chem. Commun.* **2003**, 70-71.
- (2) Abbott, A. P.; Boothby, D.; Capper, G.; Davies, D. L.; Rasheed, R. K. Deep Eutectic Solvents Formed between Choline Chloride and Carboxylic Acids: Versatile Alternatives to Ionic Liquids. *J. Am. Chem. Soc.* **2004**, *126*, 9142-9147.
- (3) Sun, H.; Li, Y.; Wu, X.; Li, G. Theoretical Study on the Structures and Properties of Mixtures of Urea and Choline Chloride. *J Mol Model* **2013**, *19*, 2433-2441.

- (4) Bahadori, L.; Abdul Manan, N. S.; Chakrabarti, M. H.; Hashim, M. A.; Mjalli, F. S.; AlNashef, I. M.; Hussain, M. A.; Low, C. T. J. The Electrochemical Behaviour of Ferrocene in Deep Eutectic Solvents Based on Quaternary Ammonium and Phosphonium Salts. *Phys. Chem. Chem. Phys.* **2013**, *15*, 1707-1714.
- (5) Zhang, Q. H.; Vigier, K. D.; Royer, S.; Jerome, F. Deep Eutectic Solvents: Syntheses, Properties and Applications. *Chem. Soc. Rev.* **2012**, *41*, 7108-7146.
- (6) Durand, E.; Lecomte, J.; Villeneuve, P. Deep Eutectic Solvents: Synthesis, Application, and Focus on Lipase-Catalyzed Reactions. *Eur. J. Lipid Sci. Technol.* **2013**, *115*, 379-385.
- (7) Carranza, A.; Pojman, J. A.; Mota-Morales, J. D. Deep-Eutectic Solvents as a Support in the Nonaqueous Synthesis of Macroporous Poly(HIPEs). *RSC Adv.* **2014**, *4*, 41584-41587.
- (8) Raghuwanshi, V. S.; Ochmann, M.; Hoell, A.; Polzer, F.; Rademann, K. Deep Eutectic Solvents for the Self-Assembly of Gold Nanoparticles: A SAXS, UV-Vis, and TEM Investigation. *Langmuir* **2014**, *30*, 6038-6046.
- (9) Russ, C.; Konig, B. Low Melting Mixtures in Organic Synthesis - an Alternative to Ionic Liquids? *Green Chem.* **2012**, *14*, 2969-2982.
- (10) Pal, M.; Rai, R.; Yadav, A.; Khanna, R.; Baker, G. A.; Pandey, S. Self-Aggregation of Sodium Dodecyl Sulfate within (Choline Chloride + Urea) Deep Eutectic Solvent. *Langmuir* **2014**, *30*, 13191-13198.
- (11) Wagle, D. V.; Zhao, H.; Baker, G. A. Deep Eutectic Solvents: Sustainable Media for Nanoscale and Functional Materials. *Acc. Chem. Res.* **2014**, *47*, 2299-2308.
- (12) Smith, E. L.; Abbott, A. P.; Ryder, K. S. Deep Eutectic Solvents (DESS) and Their Applications. *Chem. Rev.* **2014**, *114*, 11060-11082.
- (13) Chu, Q. W.; Liang, J.; Hao, J. C. Electrodeposition of Zinc-Cobalt Alloys from Choline Chloride-Urea Ionic Liquid. *Electrochim. Acta* **2014**, *115*, 499-503.
- (14) Figueiredo, M.; Gomes, C.; Costa, R.; Martins, A.; Pereira, C. M.; Silva, F. Differential Capacity of a Deep Eutectic Solvent Based on Choline Chloride and Glycerol on Solid Electrodes. *Electrochim. Acta* **2009**, *54*, 2630-2634.
- (15) Nkuku, C. A.; LeSuer, R. J. Electrochemistry in Deep Eutectic Solvents. *J. Phys. Chem. B* **2007**, *111*, 13271-13277.
- (16) You, Y.; Gu, C.; Wang, X.; Tu, J. Electrochemical Synthesis and Characterization of Ni-P Alloy Coatings from Eutectic-Based Ionic Liquid. *J. Electrochem. Soc.* **2012**, *159*, D642-D648.
- (17) Yue, D. Y.; Jia, Y. Z.; Yao, Y.; Sun, J. H.; Jing, Y. Structure and Electrochemical Behavior of Ionic Liquid Analogue Based on Choline Chloride and Urea. *Electrochim. Acta* **2012**, *65*, 30-36.
- (18) Zhang, Q. B.; Abbott, A. P.; Yang, C. Electrochemical Fabrication of Nanoporous Copper Films in Choline Chloride-Urea Deep Eutectic Solvent. *Phys. Chem. Chem. Phys.* **2015**.
- (19) Atkin, R.; Borisenko, N.; Druschler, M.; Endres, F.; Hayes, R.; Huber, B.; Roling, B. Structure and Dynamics of the Interfacial Layer between Ionic Liquids and Electrode Materials. *J. Mol. Liq.* **2014**, *192*, 44-54.
- (20) Fedorov, M. V.; Kornyshev, A. A. Ionic Liquids at Electrified Interfaces. *Chem. Rev.* **2014**, *114*, 2978-3036.
- (21) Mezger, M.; Schröder, H.; Reichert, H.; Schramm, S.; Okasinski, J. S.; Schöder, S.; Honkimäki, V.; Deutsch, M.; Ocko, B. M.; Ralston, J., et al. Molecular Layering of Fluorinated Ionic Liquids at a Charged Sapphire (0001) Surface. *Science* **2008**, *322*, 424-428.
- (22) Endres, F.; Borisenko, N.; El Abedin, S. Z.; Hayes, R.; Atkin, R. The Interface Ionic Liquid(s)/electrode(s): In Situ STM and AFM Measurements. *Faraday Discuss.* **2012**, *154*, 221-233.

- (23) Hayes, R.; Borisenko, N.; Corr, B.; Webber, G. B.; Endres, F.; Atkin, R. Effect of Dissolved LiCl on the Ionic Liquid-Au(111) Electrical Double Layer Structure. *Chem. Commun.* **2012**, *48*, 10246-10248.
- (24) Hayes, R.; Borisenko, N.; Tam, M. K.; Howlett, P. C.; Endres, F.; Atkin, R. Double Layer Structure of Ionic Liquids at the Au(111) Electrode Interface: An Atomic Force Microscopy Investigation. *J. Phys. Chem. C* **2011**, *115*, 6855-6863.
- (25) Li, H.; Endres, F.; Atkin, R. Effect of Alkyl Chain Length and Anion Species on the Interfacial Nanostructure of Ionic Liquids at the Au(111)-Ionic Liquid Interface as a Function of Potential. *Phys. Chem. Chem. Phys.* **2013**, *15*, 14624-14633.
- (26) Maolin, S.; Fuchun, Z.; Guozhong, W.; Haiping, F.; Chunlei, W.; Shimou, C.; Yi, Z.; Jun, H. Ordering Layers of [bmim][PF₆] Ionic Liquid on Graphite Surfaces: Molecular Dynamics Simulation. *J. Chem. Phys.* **2008**, *128*, 134504.
- (27) McLean, B.; Li, H.; Stefanovic, R.; Wood, R. J.; Webber, G. B.; Ueno, K.; Watanabe, M.; Warr, G. G.; Page, A.; Atkin, R. Nanostructure of [Li(G4)] TFSI and [Li(G4)] NO₃ Solvate Ionic Liquids at HOPG and Au(111) Electrode Interfaces as a Function of Potential. *Phys. Chem. Chem. Phys.* **2015**, *17*, 325-333.
- (28) Werzer, O.; Cranston, E. D.; Warr, G. G.; Atkin, R.; Rutland, M. W. Ionic Liquid Nanotribology: Mica-Silica Interactions in Ethylammonium Nitrate. *Phys. Chem. Chem. Phys.* **2012**, *14*, 5147-5152.
- (29) Li, H.; Wood, R. J.; Rutland, M. W.; Atkin, R. An Ionic Liquid Lubricant Enables Superlubricity to be "Switched on" in situ Using an Electrical Potential. *Chem. Commun.* **2014**, *50*, 4368-4370.
- (30) Costa, R.; Figueiredo, M.; Pereira, C. M.; Silva, F. Electrochemical Double Layer at the Interfaces of Hg/Choline Chloride Based Solvents. *Electrochim. Acta* **2010**, *55*, 8916-8920.
- (31) Vieira, L.; Schennach, R.; Gollas, B. In Situ PM-IRRAS of a Glassy Carbon Electrode/Deep Eutectic Solvent Interface. *Phys. Chem. Chem. Phys.* **2015**, *17*, 12870-12880.
- (32) Zhao, Y.; Truhlar, D. G. The MO6 Suite of Density Functionals for Main Group Thermochemistry, Thermochemical Kinetics, Noncovalent Interactions, Excited States, and Transition Elements: Two New Functionals and Systematic Testing of Four MO6-Class Functionals and 12 Other Functionals. *Theor. Chem. Acc.* **2008**, *120*, 215-241.
- (33) Addicoat, M. A.; Fukuoka, S.; Page, A. J.; Irle, S. Stochastic Structure Determination for Conformationally Flexible Heterogenous Molecular Clusters: Application to Ionic Liquids. *J. Comput. Chem.* **2013**, *34*, 2591-2600.
- (34) Chen, S.; Vijayaraghavan, R.; MacFarlane, D. R.; Izgorodina, E. I. Ab Initio Prediction of Proton NMR Chemical Shifts in Imidazolium Ionic Liquids. *J. Phys. Chem. B* **2013**, *117*, 3186-3197.
- (35) Shannon, R. Revised Effective Ionic Radii and Systematic Studies of Interatomic Distances in Halides and Chalcogenides. *Acta Crystallographica Section A* **1976**, *32*, 751-767.
- (36) D'Agostino, C.; Harris, R. C.; Abbott, A. P.; Gladden, L. F.; Mantle, M. D. Molecular Motion and Ion Diffusion in Choline Chloride Based Deep Eutectic Solvents Studied by ¹H Pulsed Field Gradient NMR Spectroscopy. *Phys. Chem. Chem. Phys.* **2011**, *13*, 21383-21391.
- (37) Abbott, A. P.; Harris, R. C.; Ryder, K. S. Application of Hole Theory to Define Ionic Liquids by their Transport Properties†. *J. Phys. Chem. B* **2007**, *111*, 4910-4913.
- (38) Shahbaz, K.; Baroutian, S.; Mjalli, F. S.; Hashim, M. A.; AlNashef, I. M. Densities of Ammonium and Phosphonium Based Deep Eutectic Solvents: Prediction Using Artificial Intelligence and Group Contribution Techniques. *Thermochim. Acta* **2012**, *527*, 59-66.
- (39) Gutel, T.; Santini, C. C.; Pádua, A. A. H.; Fenet, B.; Chauvin, Y.; Canongia Lopes, J. N.; Bayard, F.; Costa Gomes, M. F.; Pensado, A. S. Interaction between the π -System of

- Toluene and the Imidazolium Ring of Ionic Liquids: A Combined NMR and Molecular Simulation Study. *J. Phys. Chem. B* **2009**, *113*, 170-177.
- (40) Cao, Z.; Li, S.; Yan, T. Cation- π Interactions between a Free-Base Porphyrin and an Ionic Liquid: A Computational Study. *ChemPhysChem* **2012**, *13*, 1743-1747.
- (41) Hardacre, C.; Holbrey, J. D.; Mullan, C. L.; Nieuwenhuyzen, M.; Youngs, T. G. A.; Bowron, D. T.; Teat, S. J. Solid and Liquid Charge-transfer Complex Formation between 1-methylnaphthalene and 1-alkyl-cyanopyridinium bis{(trifluoromethyl)sulfonyl}imide Ionic Liquids. *Phys. Chem. Chem. Phys.* **2010**, *12*, 1842-1853.
- (42) Page, A. J.; Elbourne, A.; Stefanovic, R.; Addicoat, M. A.; Warr, G. G.; Voitchovsky, K.; Atkin, R. 3-Dimensional Atomic Scale Structure of the Ionic Liquid-Graphite Interface Elucidated by AM-AFM and Quantum Chemical Simulations. *Nanoscale* **2014**, *6*, 8100-8106.
- (43) Black, J. M.; Walters, D.; Labuda, A.; Feng, G.; Hillesheim, P. C.; Dai, S.; Cummings, P. T.; Kalinin, S. V.; Proksch, R.; Balke, N. Bias-Dependent Molecular-Level Structure of Electrical Double Layer in Ionic Liquid on Graphite. *Nano Letters* **2013**.
- (44) Sean, J. O. S.; Nitya, N. G.; Leonard, T. W. L.; Wulf, H. Liquid Atomic Force Microscopy: Solvation Forces, Molecular Order, and Squeeze-Out. *Japanese Journal of Applied Physics* **2010**, *49*, 08LA01.
- (45) Atkin, R.; Warr, G. G. Structure in Confined Room-Temperature Ionic Liquids. *J. Phys. Chem. C* **2007**, *111*, 5162-5168.
- (46) Horn, R. G.; Evans, D. F.; Ninham, B. W. Double-Layer and Solvation Forces Measured in a Molten-Salt and Its Mixtures with Water. *J. Phys. Chem.* **1988**, *92*, 3531-3537.
- (47) O'Shea, S. J.; Welland, M. E. Atomic Force Microscopy at Solid-Liquid Interfaces. *Langmuir* **1998**, *14*, 4186-4197.
- (48) Elbourne, A.; Voitchovsky, K.; Warr, G. G.; Atkin, R. Ion Structure Controls Ionic Liquid Near-surface and Interfacial Nanostructure. *Chem. Sci.* **2015**, *6*, 527-536.
- (49) Sweeney, J.; Webber, G. B.; Atkin, R. Near Surface Properties of Mixtures of Propylammonium Nitrate with n-alkanols 2. Nanotribology and Fluid Dynamics. *Phys. Chem. Chem. Phys.* **2015**, *17*, 26629-26637.
- (50) Bhushan, B.; Palacio, M.; Kinzig, B. AFM-based Nanotribological and Electrical Characterization of Ultrathin Wear-resistant Ionic Liquid Films. *J. Colloid Interface Sci.* **2008**, *317*, 275-287.
- (51) Fukui, K.-i.; Yokota, Y.; Imanishi, A. Local Analyses of Ionic Liquid/Solid Interfaces by Frequency Modulation Atomic Force Microscopy and Photoemission Spectroscopy. *The Chemical Record* **2014**, *14*, 964-973.
- (52) Ma, J.; Michaelides, A.; Alfè, D.; Schimka, L.; Kresse, G.; Wang, E. Adsorption and Diffusion of Water on Graphene from First Principles. *Physical Review B* **2011**, *84*, 033402.
- (53) Rangel Cortes, E.; Magana-Solis, L. F.; Arellano, J. S. Interaction of a Water Molecule with a Graphene Layer. *Revista mexicana de física* **2013**, *59*, 118-125.
- (54) Kornyshev, A. A. Double-Layer in Ionic Liquids: Paradigm Change? *J. Phys. Chem. B* **2007**, *111*, 5545-5557.
- (55) Sweeney, J.; Webber, G. B.; Rutland, M. W.; Atkin, R. Effect of Ion Structure on Nanoscale Friction in Protic Ionic Liquids. *Phys. Chem. Chem. Phys.* **2014**, *16*, 16651-16658.
- (56) Steiner, T. The Hydrogen Bond in the Solid State. *Angew Chem Int Ed* **2002**, *41*, 48-76.
- (57) Sedev, R. Electrowetting: Electrocapillarity, Saturation, and Dynamics. *Eur. Phys. J. Spec. Top.* **2011**, *197*, 307-319.
- (58) Paneru, M.; Priest, C.; Sedev, R.; Ralston, J. Static and Dynamic Electrowetting of an Ionic Liquid in a Solid/Liquid/Liquid System. *J. Am. Chem. Soc.* **2010**, *132*, 8301-8308.
- (59) Wang, S.; Zhang, Y.; Abidi, N.; Cabrales, L. Wettability and Surface Free Energy of Graphene Films. *Langmuir* **2009**, *25*, 11078-11081.

TOC

